The documentation and process conversion measures necessary to comply with this document shall be completed by 9 October 2020.

INCH-POUND

MIL-PRF-19500/623F
10 July 2020
SUPERSEDING
MIL-PRF-19500/623E
23 July 2015

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, DARLINGTON, PNP SILICON,
HIGH-POWER, DEVICE TYPE 2N7371,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon, high-power Darlington transistor. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device.

1.2 Package outlines. The device package outline is a three terminal flange mount header TO-254AA package in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25 °C.

Type	P_T (1) T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$ (2)
	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N7371	100	-100	-100	-5.0	-0.2	-12	−65 to +200	1.5

(1) See figure 2 for temperature-power derating curves.
(2) See figure 3, transient thermal impedance graph.

1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25 °C.

Limit	h_{FE1} (1) V_{CE} = -3.0 V dc I_C = -6.0 A dc	$V_{BE(SAT)}$ (1) I_C = -12.0 A dc I_B = -120 mA dc	$V_{CE(SAT)}$ (1) I_C = -12.0 A dc I_B = -120 mA dc	$\lvert h_{fe} \rvert$ V_{CE} = -3.0 V dc I_C = -5.0 A dc f = 1 MHz
		V dc	V dc	
Min	1,000			10
Max	18,000	-4.0	-3.0	250

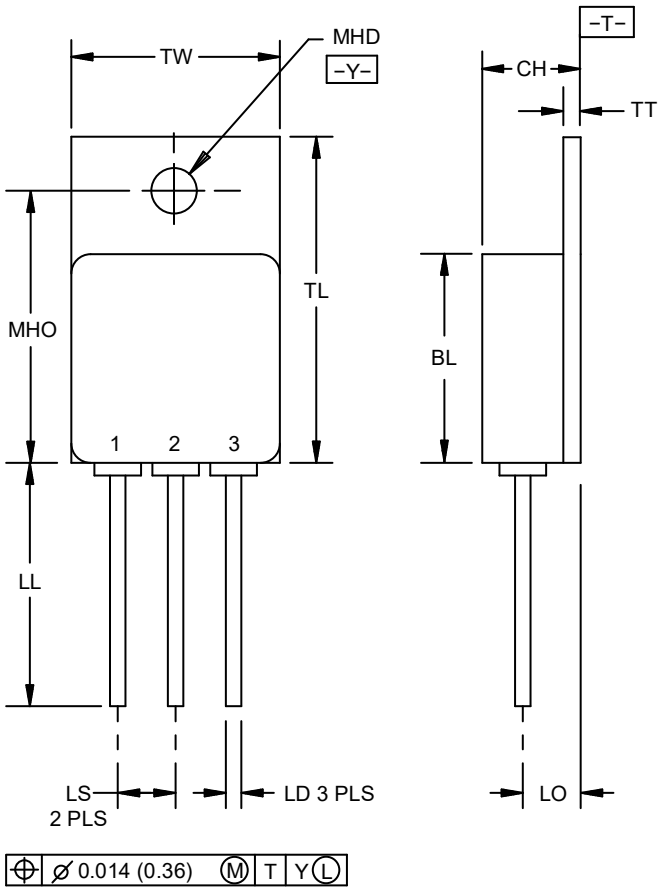
(1) Pulsed see 4.5.1.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.94	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84
Term 1	Base			
Term 2	Collector			
Term 3	Emitter			

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Protrusion of ceramic eyelets included in dimension LL.
3. All terminals are isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to Øx symbology.

FIGURE 1. Dimensions and configuration of three terminal flange mount header TO-254AA package.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are "7371".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2).

3.4.2 Lead formation. Where a choice of lead formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100 percent dc testing in accordance with subgroup 2 of table I herein on all devices that had its leads altered.

3.4.3 Lead isolation. Methods used for electrical isolation of the terminal feed through shall employ materials that contain a minimum of 90 percent ceramic AL_2O_3 or equivalent. Examples of such construction techniques are metalized ceramic eyelets or ceramic walled packages.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II.).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTX and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CEX1} and h_{FE1}	I_{CEX1} and h_{FE1}
11	Subgroup 2 of table I herein; I_{CEX1} and h_{FE1}; ΔI_{CEX1} = 100 percent of initial value or -2 µA dc, whichever is greater. Δh_{FE1} = ±40 percent of initial value.	I_{CEX1} and h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; I_{CEX1} and h_{FE1}; ΔI_{CEX1} = 100 percent of initial value or -2 µA dc, whichever is greater. Δh_{FE1} = ±40 percent of initial value.	Subgroup 2 of table I herein; I_{CEX1} and h_{FE1}; ΔI_{CEX1} = 100 percent of initial value or -2 µA dc, whichever is greater. Δh_{FE1} = ±40 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a. Quality levels JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: T_J = +175 °C min, V_{CE} = -10 to -30 V dc, T_A = +30 ±5 °C.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, and t_{MD} (and V_C where appropriate). The $Z_{\theta JX}$ limit used in screen 3c shall comply with the thermal impedance graph on figure 3 (less than or equal to the curve value at the same t_H time) and/or shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	$V_{CB} \geq$ -10 V dc.
B5	2037	Bond strength, test condition D.
B6	3131	See 4.5.2.

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4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1037	V_{CB} ≥ -10 V dc.
B5	3131	See 4.5.2.
B6	1032	T_A = +200 °C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	Test condition B.
C2	2036	Test condition A, weight = 4.5 kg, t = 15 seconds.
C6	1037	V_{CB} ≥ -10 V dc.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be -1.0 A dc.

b. Collector to emitter voltage magnitude shall be ≥ -10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25 °C ≤ T_R ≤ +75 °C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 1.5 °C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3131	See 4.3.2	$Z_{\theta JC}$			
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = -100 mA dc, pulsed (see 4.5.1)	$V_{CEO(sus)}$	-100		V dc
Collector - emitter cutoff current	3036	Bias condition D; V_{CE} = -50 V dc	I_{CEO}		-1.0	mA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -5 V dc	I_{EBO}		-2.0	mA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{BE} = +1.5 V dc, V_{CE} = -100 V dc	I_{CEX1}		-20	µA dc
Base - emitter saturated voltage	3066	Test condition A; I_C = -12 A dc, I_B = -120 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)}$		-4.0	V dc
Collector - emitter saturated voltage	3071	I_C = -12 A dc; I_B = -120 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)}$		-3.0	V dc
Forward - current transfer ratio	3076	V_{CE} = -3.0 V dc; I_C = -6.0 A dc, pulsed (see 4.5.1)	h_{FE1}	1000	18,000	
Forward - current transfer ratio	3076	V_{CE} = -3.0 V dc; I_C = -12 A dc, pulsed (see 4.5.1)	h_{FE2}	150		
Subgroup 3						
High temperature operation:		T_A = +150 °C				
Collector to emitter cutoff current	3041	Bias condition C; V_{BE} = +1.5 V dc, V_{CE} = -100 V dc	I_{CEX2}		-1.0	mA dc
Low temperature operation:		T_A = −55 °C				
Forward - current transfer ratio	3076	V_{CE} = -3.0 V dc; I_C = -6.0 A dc, pulsed (see 4.5.1)	h_{FE3}	300		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching parameters						
Turn-on		See figure 4	t_{on}		2.0	µs
Turn-off		See figure 4	t_{off}		10	µs
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -3.0 V dc, I_C = -5.0 A dc, f = 1 MHz	|hfe|	10	250	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25 °C; t ≥ 1 s; 1 cycle; (see figure 5)				
Test 1		V_{CE} = -8.3 V dc; I_C = -12.0 A dc				
Test 2		V_{CE} = -30 V dc; I_C = -3.3 A dc				
Test 3		V_{CE} = -90 V dc; I_C = -150 mA dc				
Safe operating area (clamped inductive)	3053	Load condition B (clamped inductive load); T_A = +25°C; tr + tf ≤ 1.0 µs; duty cycle ≤ 2 percent; tp = 1 ms; (vary to obtain I_C); Rs = 0.10 ohms; R_{BB1} = 80 ohms; V_{BB1} = -16 V dc; R_{BB2} = 100 ohms; V_{BB2} = -1.5 V dc; I_C = -12 A dc; V_{CC} = -20 V dc; R_L ≤ 2 ohms; L = 10 mH; (Stancor C-2688 or equivalent) clamp voltage = -100 +0, -5 V dc; Device fails if clamp voltage not reached.				
Electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

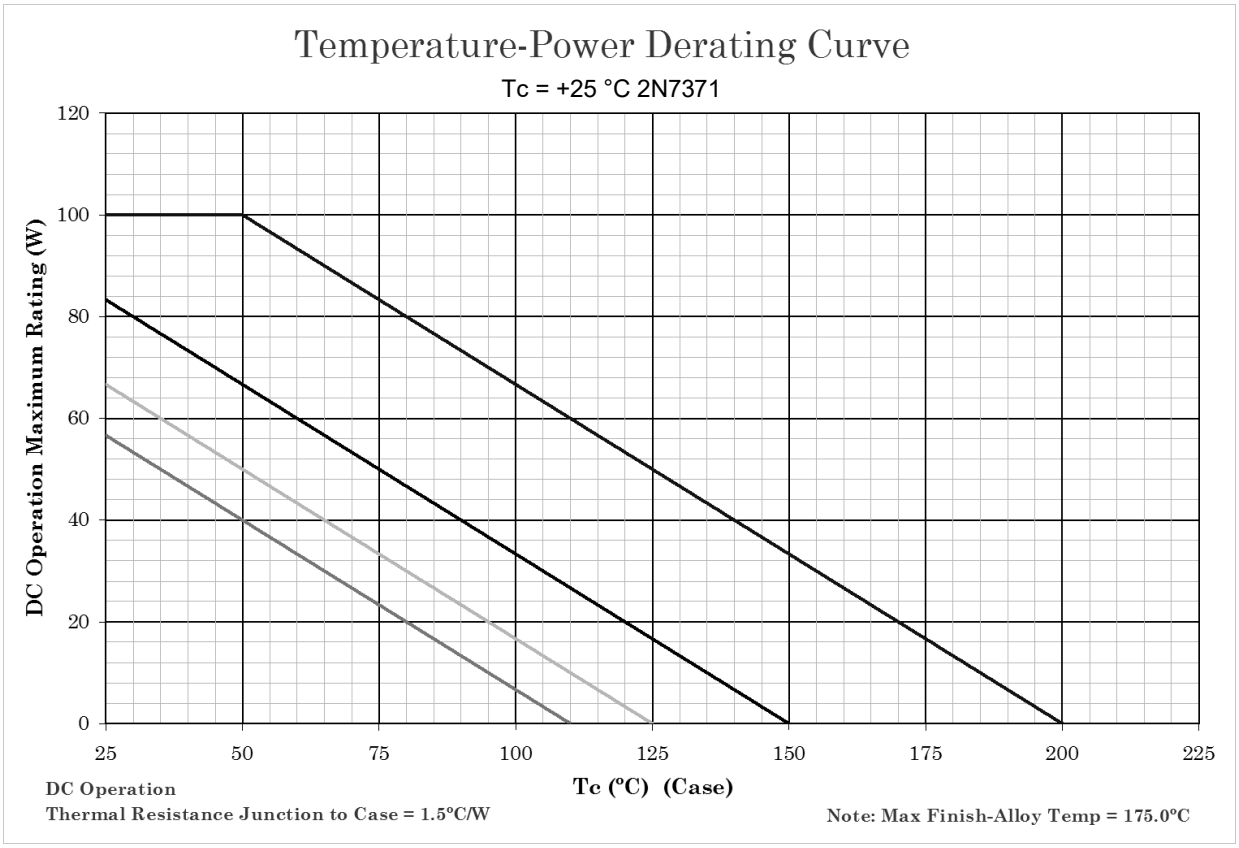
1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling (air to air)	1051	500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
High temperature reverse bias	1039	Condition A; 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		See table E-IX of MIL-PRF-19500, group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition B for devices < 400 V.	

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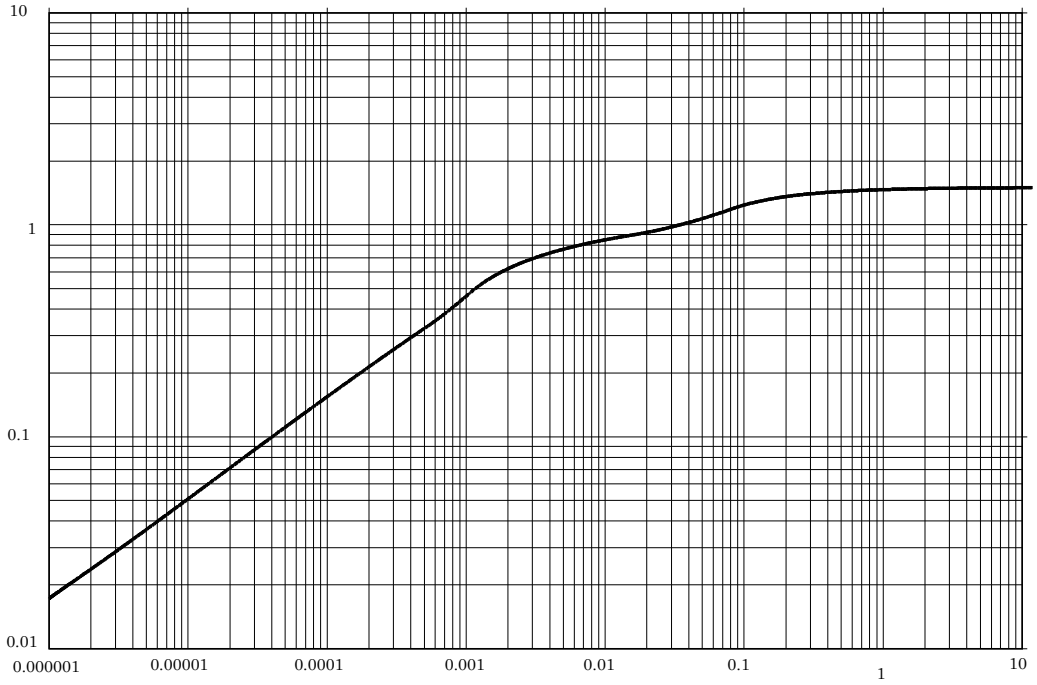
NOTES:

1. Maximum theoretical derate design curve. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified (see 1.3).
3. Derate design curve chosen at T_J ≤ +150 °C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125 °C, and +110 °C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Temperature-power derating graph.

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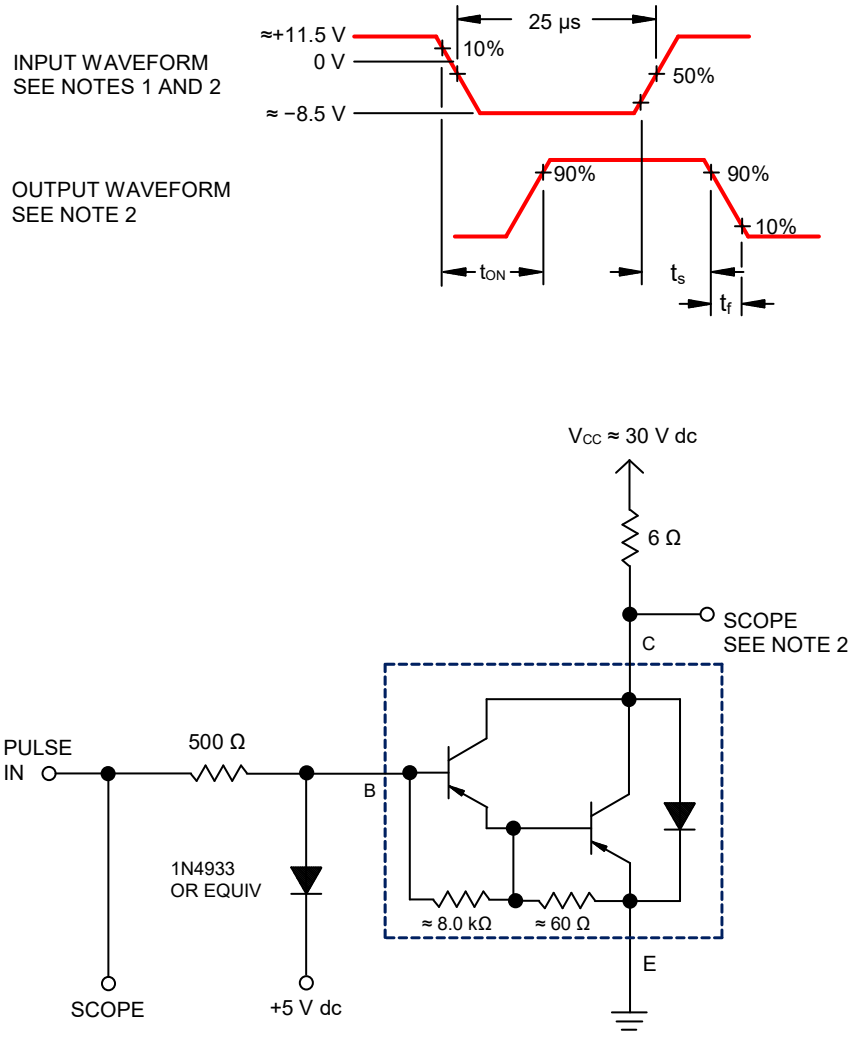
Maximum Thermal Impedance



T_C = +25 °C. Thermal resistance = 1.5 °C/W.

FIGURE 3. Transient thermal impedance graph.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20$ ns, $t_f \leq 20$ ns, $Z_{OUT} = 50\ \Omega$, PW = 25 µs, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 20$ ns, $Z_{IN} \geq 20\ k\Omega$, $C_{IN} \leq 11.5$ pF.
3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 4. Pulse response test circuit.

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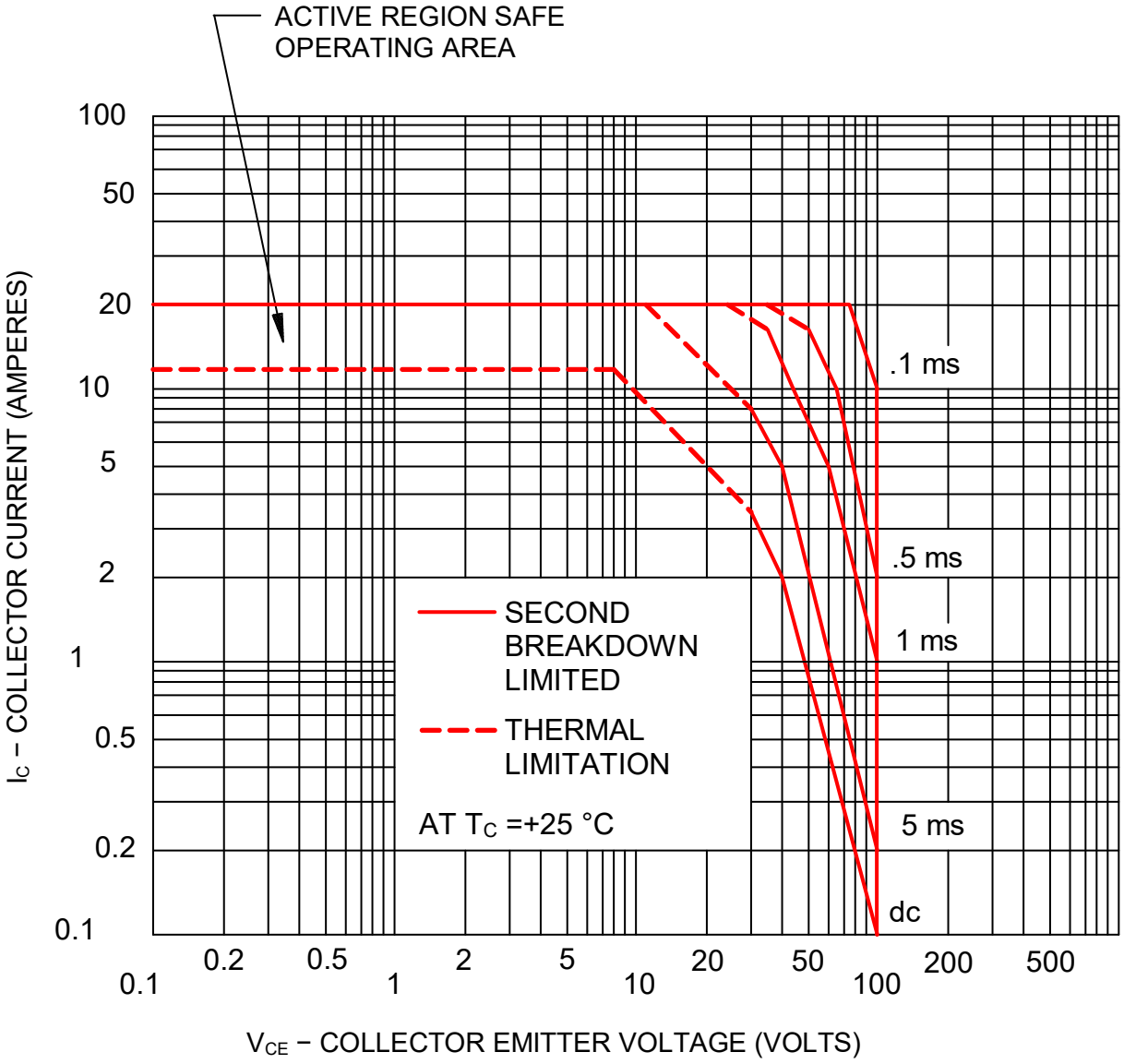


FIGURE 5. Safe operating area.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

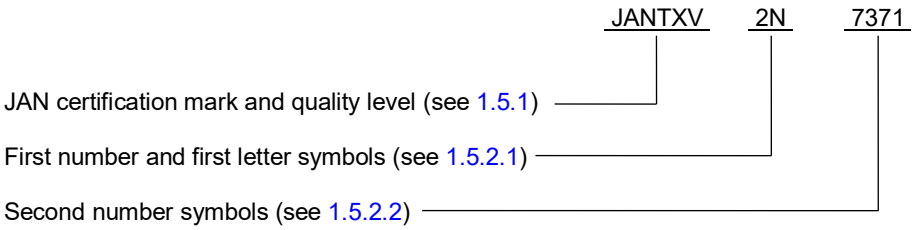
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1) and if applicable, any needed lead formation (see 3.4.2).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML−19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N7371	JANTX2N7371	JANTXV2N7371	JANS2N7371

6.6 Interchangeability information. The die of device type 2N7371 covered by this specification sheet is the same die as device type 2N3792 covered by MIL-PRF-19500/501. However, the package outline of this specification sheet is a TO-254AA package whereas the package outline of MIL-PRF-19500/501 is a TO-204AA (formerly TO-3) package. The devices from this specification sheet are preferred over the devices of MIL-PRF-19500/501 whenever interchangeability is not a problem.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
Navy − AS
Air Force − 19

Preparing activity:
DLA − CC

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